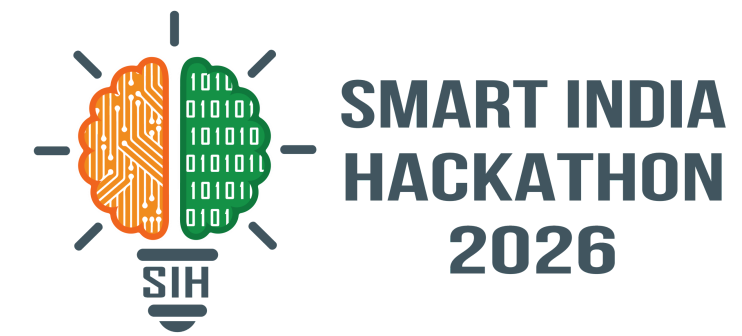


SMART INDIA HACKATHON 2026



Problem Statement ID : SIH26073

Problem Statement Title : AI/ML-Based Intelligent Anomaly Detection for Automatic Weather Stations (AWS)

Theme : Software

Problem Statment Category : Disaster Management

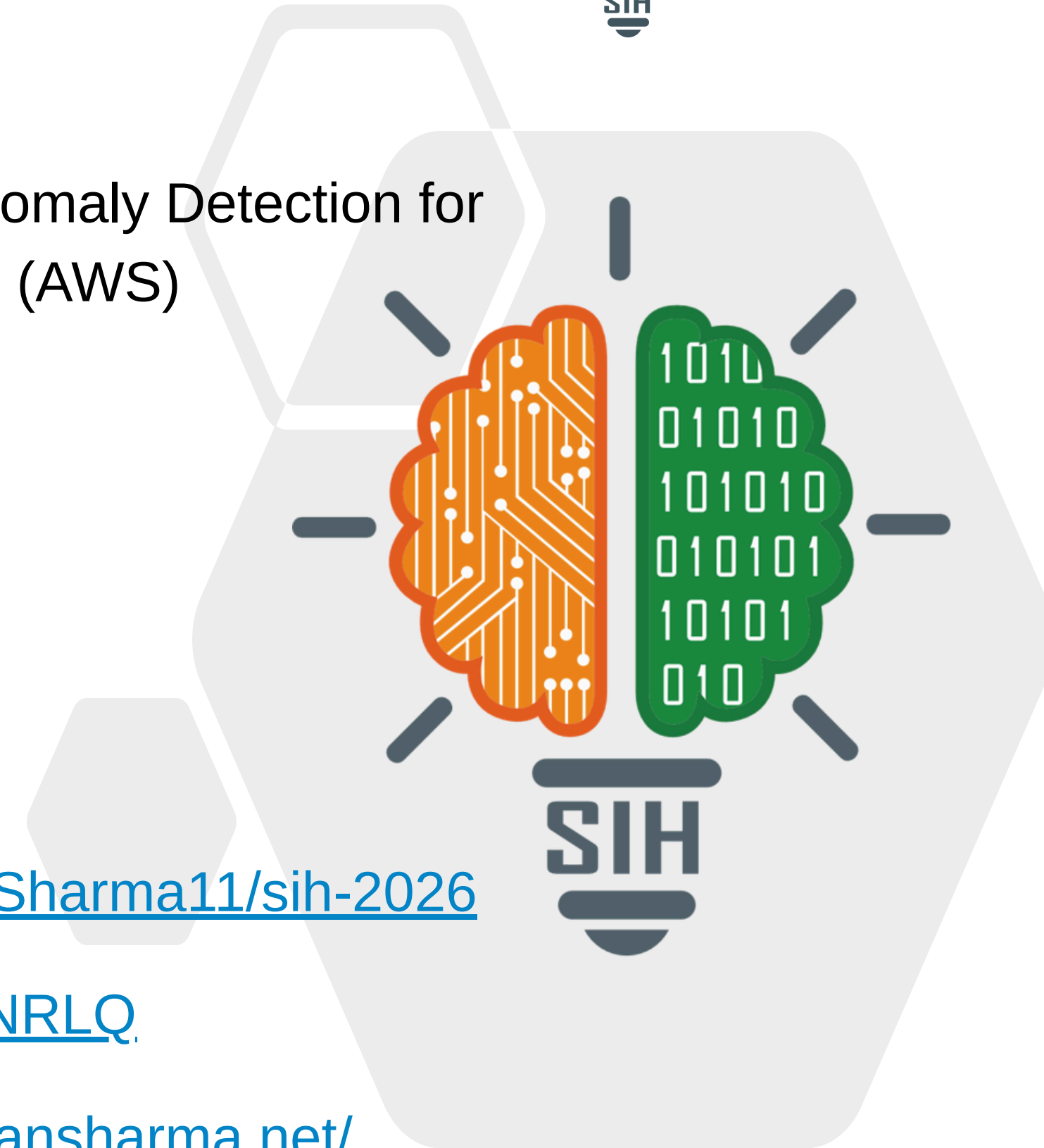
Team ID : 178908

Team Name : __Think Tank__

Github Repository Link : <https://github.com/RoshanSharma11/sih-2026>

Youtube Video Link : <https://youtu.be/emdL6D2NRLQ>

Prototype Demo Link : <https://www.skyguard.roshansharma.net/>



SkyGuard AI – An intelligent, real-time quality control platform for Automatic Weather Stations (AWS) that detects, **explains and classifies anomalies** using only temperature, pressure and humidity data.



CHALLENGE



Sensor faults, spikes and frozen values



Difficult to distinguish faults from real weather events (e.g., 55°C spike)



Manual verification is time-consuming and non-scalable



OUR SOLUTION – SKYGUARD AI



Data Ingestion

24h window (T, P, H) + nearby stations (≤ 100 km)



Intelligent Detection

Rule-based QC + LSTM + Spatial Consensus



Anomaly Classification

5-way verdict with confidence and explanation



Corrected Estimation

Suggest last hour values when sensor is untrusted



Dashboard & Alerts

Real-time visualization, reasoning and maintenance hints



OUTCOME



Reliable, explainable and real-time anomaly detection



Reliable, explainable and real-time anomaly detection



Actionable alerts with corrected values (when needed)



Scalable across large AWS network



HOW IT ADDRESSES THE PROBLEM

Spikes and abnormal readings (e.g., 55°C)



Hybrid detection (rules + ML + neighbour check) identifies hardware faults vs real events

Frozen sensors & communication errors



Physical rules and temporal analysis detect stuck/missing values

Difficult to separate weather events from sensor faults



Spatial consensus using nearby stations (CW-IDW) to verify if anomaly is regional or isolated

Lack of explainability and actionable insights



Provides confidence, reason, contributing factors and maintenance hints



INNOVATION & UNIQUENESS

01



Physics-informed Anomaly Detection

LSTM autoencoder trained with T-P-H physics (Magnus dew-point) for climate-aware detection

02



Spatial Consensus (CW-IDW)

Uses correlation-weighted inverse-distance to distinguish genuine weather events from hardware anomalies

03



Root-cause Classification

5-way verdict with confidence and human-readable explanation for each alert

04



Intelligent Correction

Suggests physically consistent last hour values (Gaussian overlay) when sensor is untrusted

1. OVERALL SYSTEM FLOW

AWS / IMD Data

- T, RH, P (hourly)
- IMD API (real-time)

Data Ingestion & Preprocessing

- Handle missing values
- Data cleaning

QC / Anomaly Detection

- Physical rules
- LSTM autoencoder

Classification & Correction

- 5-way classification
- Last-hour repair

Result Storage

- Store results
- SQLite (local)

User Interface

- Dashboard & alerts
- Manual review

2. TECHNOLOGY STACK

Python

PyTorch

NumPy

Pandas

FastAPI

SQLAlchemy

SQLite

Streamlit

Matplotlib

Meteostat

IMD API

Docker

AWS EC 2

3. METHODOLOGY (DETAILED)

01

Physical Rules & Domain Constraints.

Range, step and consistency checks (Magnus Td, RH box, Clausius-Clapeyron).

02

PIML LSTM Autoencoder (Temporal)

Learn normal T/P/H behaviour for anomaly detection and reconstruction.

03

Spatial Buddy Check (IDW)

Learn normal T/P/H behaviour for anomaly detection and reconstruction.

04

Anomaly Classification & Correction

5-way classification with last-hour reconstruction when required.

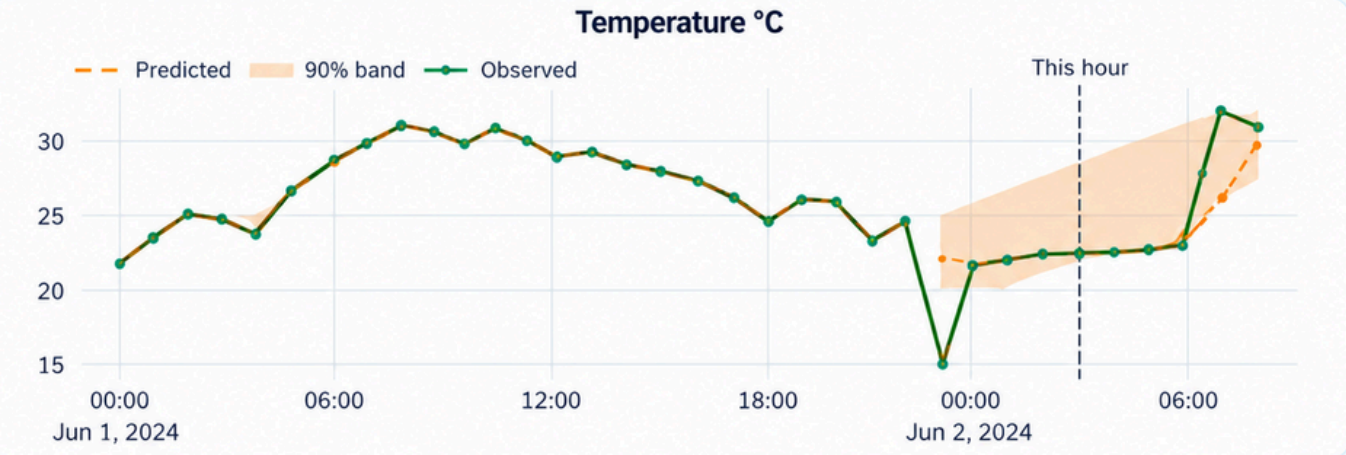
05

Confidence & Explainability

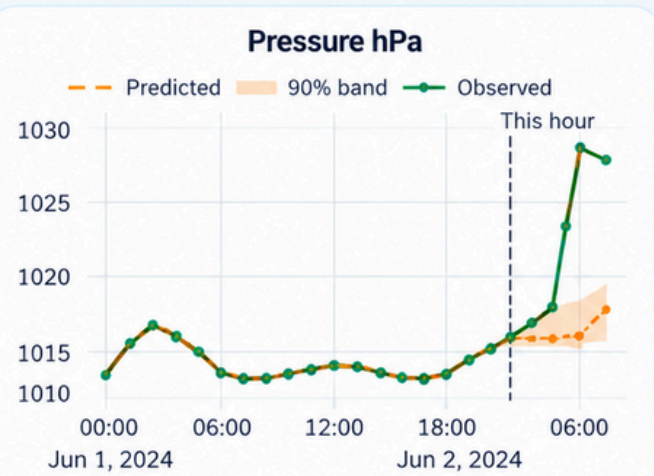
5-way classification with last-hour reconstruction when required.

4. WORKING PROTOTYPE (PREDICTION)

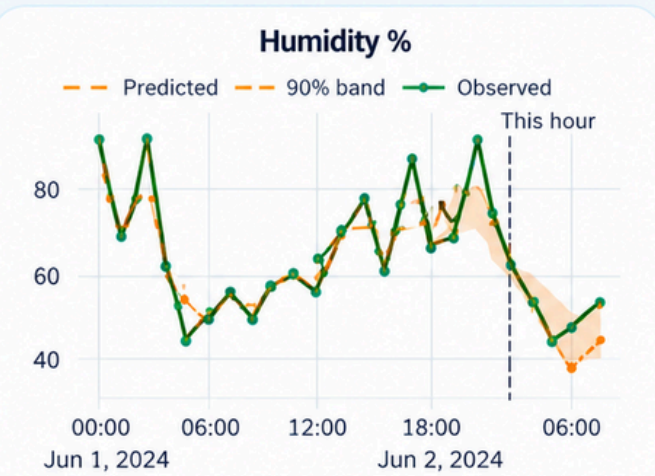
Temperature °C



Pressure hPa



Humidity %

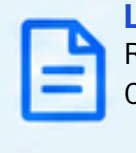


LAST 24 VERDICTS



5. MODEL DETAILS AND VALIDATION (SKYGUARD V2)

 **PIML LSTM Autoencoder**
Input 24 x 3 → LSTM 64 → latent 32 → 24 x 3
45,027 parameters

 **Loss Function**
Reconstruction loss + 0.1 x physics loss

 **Best Validation MSE**
0.001369 (epoch 36)

 **Clean Validation p99 MSE**
0.00618

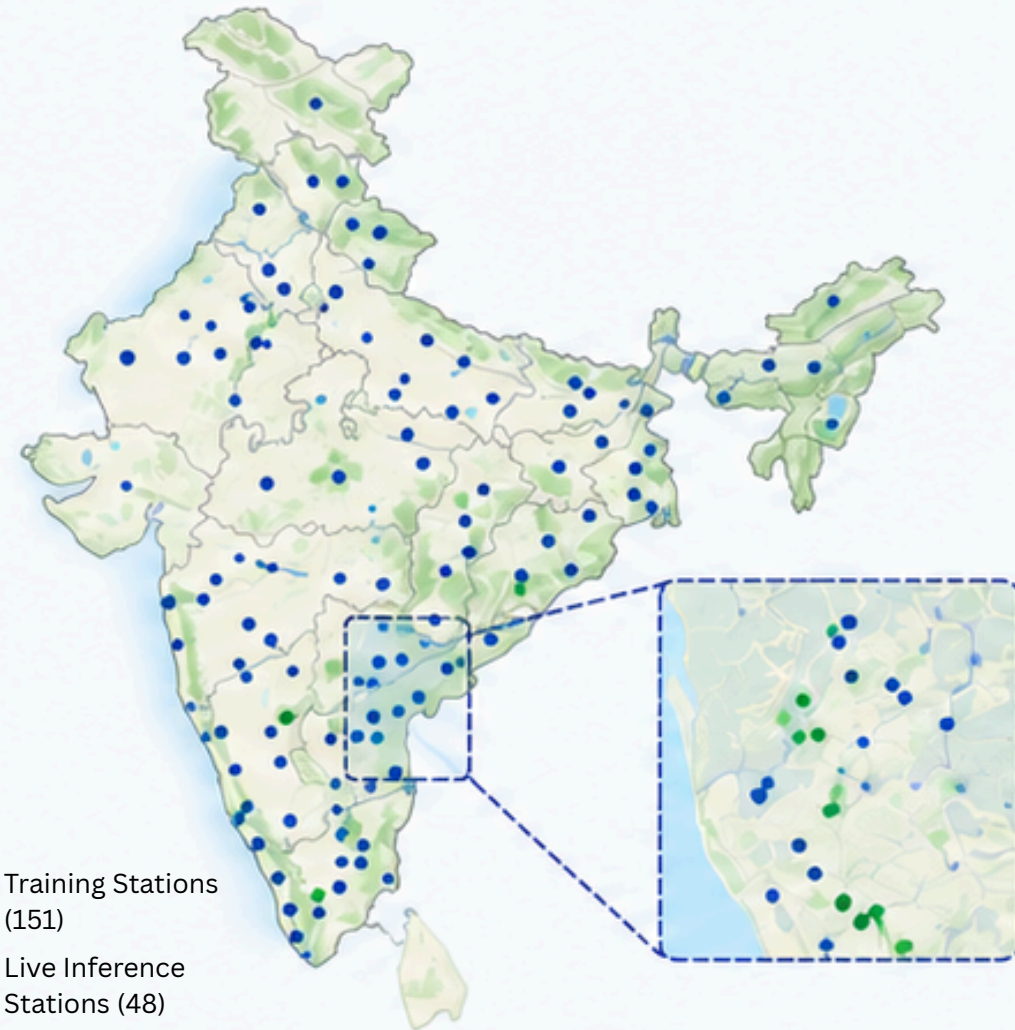
 **2023 Gate Results (not 2024)**
Temp MAE 0.0175 | RH MAE 0.0345 | P MAE 0.0106

 **Slice B GAT Threshold**
0.875 | Hardware Agree 0.998 | Weather Agree 0.998 | Clean Agree 0.966

3



DEPLOYMENT SCALE & COVERAGE



151
AWS Stations
(Training Network)



48
Live IMD mapped stations
(Inference)



2020 - 2022
Training Data
(151 stations)



2023
Validation Data
(Threshold Tuning)



2024
Test Data
(never used for tuning)



POTENTIAL CHALLENGES & RISKS



Incomplete or missing data from some stations



Rare and unseen fault patterns



False alarms during extreme weather events



Real-Time API access and API Downtime



Scalability for large station networks



MITIGATION STRATEGY



Robust preprocessing, missing value handling and spatial backup from nearby stations.



Hybrid approach (rules + ML + spatial analysis) for better generalization



Neighbouring stations (CW-IDW) and multivariate consistency checks (T, P, H)



Caches, retry mechanisms and fallback to historical data



Modular, containerized architecture deployable on cloud (AWS EC2)



FEASIBILITY & VIABILITY ANALYSIS



Uses existing technologies & open data



Historical data available
(Meteostat & IMD API)



Low infrastructure requirements
(CPU-based inference)



Real-time capable
(low latency)



Scalable to large network of stations
(151+ stations)



Cost-efficient deployment
(AWS EC2)



Modular and maintainable architecture



DEPLOYMENT ROADMAP



Prototype
(Working Demo)



Pilot Deployment
(Selected Stations)



Full-scale Deployment
(PAN India)

 IMPACT AREAS**DISASTER
PREPAREDNESS**

- Early detection of sensor faults
- More reliable alerts during extreme weather
- saves lives and property

**AVIATION &
TRANSPORT**

- Trustworthy weather data for operational safety
- Fewer disruptions during faulty sensors
- Smoother and safer options

**AGRICULTURE**

- Reliable local weather data
- Better crop planning and irrigation
- Reduces economic losses for farmers

**RESEARCH & CLIMATE
CHANGES**

- High-quality cleaned data
- Supports climate change and modelling
- Better understanding of long-term trends

 TRIPLE IMPACT**SOCIAL IMPACT**

- Safer communities through reliable alerts
- Supports disaster management
- Builds trust in public weather services

**ECONOMIC IMPACT**

- Reduces false-alert losses
- Improves sector efficiency
- Cost-effective & scalable

**ENVIRONMENTAL IMPACT**

- Better monitoring of extreme weather events
- Supports climate resilience and adaptation
- Helps in sustainable resource management

 TARGET AUDIENCE

Government agencies



Meteorological & Research Institutions



Infrastructure & Critical Sectors



General Public & Local Communities

 REAL-WORLD VALUE**Higher Safety***Protects people and critical infrastructure***Greater Efficiency***Reduces losses and improves outcomes across sectors***Climate Resilience***Strengthens long-term adaptation and planning*



RESEARCH FOUNDATION

- 01

**Domain Knowledge**
AWS observation standards and quality control
- 02

**ML Method**
LSTM reconstruction for anomaly detection
- 03

**Spatial Intelligence**
Graph Attention Networks and neighbor consistency
- 04

**Data Source**
Historical weather observations



Grounded in established meteorological QC, anomaly detection and spatial validation



KEY REFERENCES

WMO	 WMO-No. 8 — AWS quality control World Meteorological Organization. Guide to Instruments and Methods of Observation (WMO-No. 8), 2024 ed.	 Used for: range / step / missing / freeze and the neighbor (buddy) check.	 https://library.wmo.int/viewer/68695/
MAGNUS	 Magnus dew point Alduchov, O. A. & Eskridge, R. E. (1996). Improved Magnus form approximation of saturation vapor pressure. Journal of Applied Meteorology, 35(4), 601–609.	 Used for: Td(T, RH) as a hard fail if Td > T, and the same formula in the LSTM physics loss (a = 17.27, b = 237.7).	 https://doi.org/10.1175/1520-0450(1996)035%3C0601:IMFAOS%3E2.0.CO;2
LSTM	 LSTM autoencoder anomaly detection Malhotra, P. et al. (2016). LSTM-based Encoder-Decoder for Multi-sensor Anomaly Detection. ICML Anomaly Detection Workshop. arXiv:1607.00148.	 Used for: 24×3 reconstruction autoencoder; last-hour-weighted MSE vs a frozen 2023 threshold.	 https://arxiv.org/abs/1607.00148
GAT	 Graph Attention Networks Veličković, P. et al. (2018). Graph Attention Networks. ICLR.	 Used for: spatial head on the 100 km station graph (demo stories 2 vs 3). Ingest uses CW-IDW.	 https://arxiv.org/abs/1710.10903
METEOSTAT	 Meteostat hourly stations Meteostat. Hourly weather-station observations (T, RH, P).	 Used for: 151-station 2020–2023 train/val; 2024 held out; live map is the 48 IMD-mapped stations.	 https://dev.meteostat.net/